

you can easily link devices in the cloud, on-premises, and at the edge.

The fact that it is open source means that there is a community working on improving it. Businesses in need of expert assistance may find comfort in the fact that NetFoundry offers commercial support as well.

Let's now see how we can set up a Zero Trust network environment using OpenZiti.

Prerequisites and installation

Before you begin, ensure you have the following:

- A server or cloud instance to host the OpenZiti controller.
- Basic knowledge of command-line interface (CLI) operations.
- Installed dependencies such as Go (version 1.16 or later) and Docker.

Open your terminal and run the following command to clone the OpenZiti repository:

```
git clone https://github.com/openziti/ziti.git
```

```
cd ziti
```

Now navigate to the cloned directory and build the necessary components:

```
make all
```

Setting up the controller

Use Docker to run the OpenZiti controller. Execute the following command:

```
docker-compose up -d
```

Check the status of the controller by running it. You should see the OpenZiti controller listed among the running containers.

Configure the network

Open your web browser and navigate to <http://localhost:1280> to access the OpenZiti dashboard.

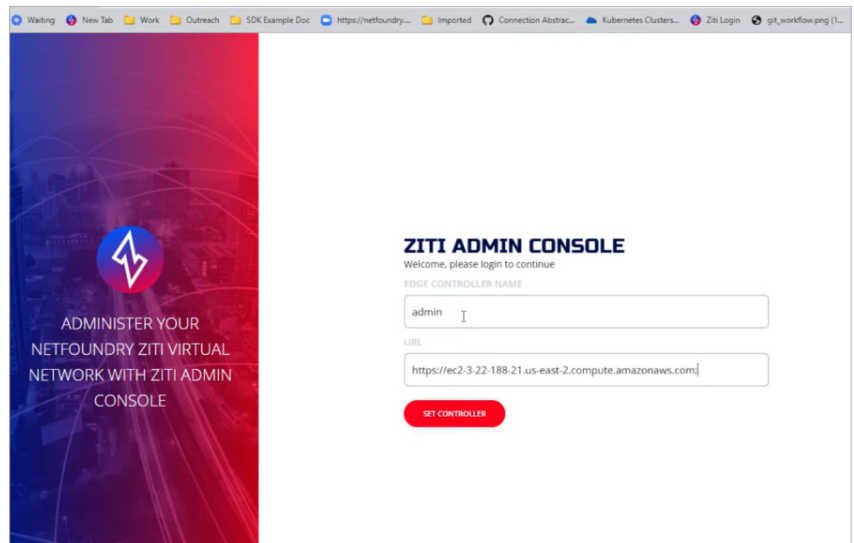


Figure 2: Admin UI running on localhost

Create a new network as follows:

- Click on the 'Networks' tab.
- Select 'Create Network' and fill in the required details such as network name and description.

Define the services that will be accessible through the OpenZiti network. Click on the 'Services' tab and add your services accordingly.

Install the edge router

Use the following command to deploy the edge router:

```
docker run -d --name ziti-edge-router -p 8443:8443 openziti/ziti-edge-router
```

Access the edge router configuration and set it up to connect to your network.

Client configuration

Depending on your programming language, install the appropriate OpenZiti SDK. For example, for Go, you can run:

```
go get github.com/openziti/ziti-sdk-golang
```

Develop your client application using the OpenZiti SDK. Ensure that it connects to the OpenZiti network and utilises the services you configured.

Testing the setup

Execute your client application and verify that it can successfully connect to the services through the OpenZiti network.

Use the OpenZiti dashboard to monitor network traffic and ensure everything is functioning as expected.

Access the controller's admin UI at <http://localhost:1280> (credentials were set during controller setup).

This detailed method will help create a Zero Trust network environment via the OpenZiti setup and configuration process. Secure management and monitoring of every part of your network can be guaranteed by following these steps—deploying the controller, setting up the edge routers, enrolling client endpoints, and creating granular network policies.

Happy networking! 🐧

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